



CHEMISTRY (Periodic Table)

PERIODIC TABLE – a table listing the elements in increasing atomic numbers.

- Horizontal rows = Periods
- Vertical columns = Groups or families

Periodic Law: The properties of the elements recur periodically when the elements are arranged according to their increasing order of their atomic numbers.

Groups: Numbered with roman numerals; elements in each group have the same number of electrons in their outermost shells, so they all behave similarly.

Atomic Radius: Presumably determined the size of an atom in each element.

Ionization Energy: Energy required to remove an outermost electron from an isolated gaseous atom; the HIGHER the ionization energy = HARDER to remove an electron.

Electron Affinity: Energy change when an electron is gained by an ion; the GREATER affinity for an added electron = GREATER energy released.

METALLIC PROPERTIES:

- ☐ Solid at room temperature (except for Mercury, which is liquid)
- ☐ High melting point
- ☐ Ionization of energy is LOW
- ☐ Effective conductors
- ☐ Luster (reflects light)
- ☐ Ductile (can be drawn into wires)
- ☐ Can be pounded into SHEETS (Malleable)

CHEMICAL REACTIONS – represented by chemical equations



REACTANTS: starting materials (compounds before the arrows)

PRODUCTS: new compounds formed (compounds after the arrow)

The equation states that: 2 molecules of ethane C_3H_6 is needed to react with 7 molecules of Oxygen O_2 to form 2 molecules of CO_2 and 3 molecules of H_2O

THE LAW OF CONSERVATION OF MASS: In a chemical reaction, mass of products EQUALS mass of reactants.

TYPES OF CHEMICAL REACTIONS:

- 1. Combination Synthesis:** involves combination of two or more substances to produce a single complex compound. ($A + B \rightarrow AB$)
 - $8Fe + S_8 \rightarrow 8FeS$
- 2. Decomposition:** a compound is broken into smaller chemical species. ($AB \rightarrow A + B$)
 - $2H_2O \rightarrow 2H_2 + O_2$
- 3. Single Displacement/ Substitution:** one element is displaced from a compound by another element. ($A + BC \rightarrow AC + B$)
 - $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
- 4. Double Displacement/ Metathesis:** two compounds exchange bonds or ions in order to form different compounds ($AB + CD \rightarrow AD + CB$)
 - $NaCl(aq) + AgNO_3(aq) \rightarrow NaNO_3(aq) + AgCl(s)$
- 5. Acid – base:** type of double displacement reaction that occurs between an acid and a base. The H^+ ion in the acid reacts with the OH^- ion in the base to form water and an ionic salt. ($HA + BOH \rightarrow H_2O + BA$)
 - $HBr + NaOH \rightarrow NaBr + H_2O$
- 6. Combustion:** oxygen combines with another compound to form Carbon dioxide and water.
 - $C_{10}H_8 + 12O_2 \rightarrow 10CO_2 + 4H_2O$

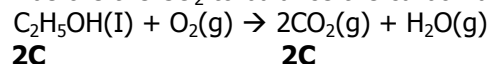
BALANCING CHEMICAL EQUATIONS

STEP 1: Most chemical equations can be balanced by trial and error. Start with the most complicated molecules (those containing the greatest number of atoms).

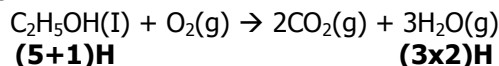
- **Example:** $C_2H_5OH(l) + O_2(g) \rightarrow CO_2(g) + H_2O(g)$



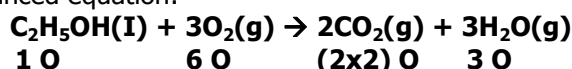
STEP 2: The most complicated here is C_2H_5OH . Begin by balancing the products that contain the atoms in C_2H_5OH . Since C_2H_5OH contains two carbon atoms, we place a 2 before the CO_2 to balance the carbon atoms.



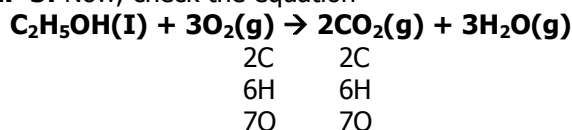
STEP 3: Since C_2H_5OH contains six hydrogen atoms, the hydrogen atoms can be balanced by placing a 3 before the H_2O .



STEP 4: Last, we balance the oxygen atoms. Note that the right side of the equation contains seven oxygen atoms, while the left side has only three. We can correct this by putting a 3 before the O_2 to produce the balanced equation:



STEP 5: Now, check the equation



Factors Affecting Rate of Chemical Reactions:

1. Concentration of Reactants

- More molecules available = faster the reactions

2. Surface Area

- Greater surface area = faster reaction

3. Temperature

- Greater temperature (greater kinetic energy) = faster reaction

4. Catalyst

- Substances that sped up rate of reaction without being consumed in the reaction.

BASIC STOICHIOMETRY

- Mole (mol)** – quantity substance that contains Avogadro's number of units or particles (Avogadro's number = 6.02×10^{23})
- Molar mass (MM)** – mass in grams of one mole of the substance; for elements it is equal to the atomic mass; for compounds it is equal to the molecular mass
- Molecular mass / Formula mass (ama)** – sum of the atomic masses of all the constituent atoms in a molecular compound.

REMEMBER:

- Avogadro's Number = 6.02×10^{23}
- No. of atoms = No. of moles x Avogadro's no.
- No. of moles = Given mass/ molar mass

Example:

- How many atoms are in 5 moles of Carbon (C)?

$$\begin{aligned} \bullet \text{ No. of atoms of C} &= 5 \text{ moles C} \times \frac{6.02 \times 10^{23}}{1 \text{ mole C}} \\ &= 3.01 \times 10^{24} \text{ atoms} \end{aligned}$$

- How many moles of sodium (Na) are there is 64g of sodium? Molar mass of Na = 32g/ mol

$$\begin{aligned} \bullet \text{ No. of moles of Na} &= 64g \text{ Na} \times \frac{1 \text{ mole Na}}{32g \text{ Na}} \\ &= 2 \text{ moles Na} \end{aligned}$$

- What is the mass of 2 mol $NaHCO_3$?

$$\bullet \text{ Na: } 1 \times 23 = 23$$

$$\text{H: } 1 \times 1 = 1$$

$$\text{C: } 1 \times 12 = 12$$

$$\text{O: } 3 \times 16 = 48$$

$$84.00 \text{ g/ mol}$$

$$\text{Mass} = \text{No. of moles} \times \text{molar mass}$$

$$= 2 \text{ mol} \times 84g/ \text{mol}$$

$$= 168g \text{ NaHCO}_3$$

GAS LAWS

- Charles's Law** – The volume of a gas is directly proportional to the temperature while keeping the amount of gas and its pressure constant.

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

- Boyle's Law** – At constant temperature, the volume of a fixed amount of gas is inversely proportional to pressure.

$$P_1V_1 = P_2V_2$$

- Gay-Lussac's Law** – The pressure of a fixed amount of gas is directly proportional to the temperature while keeping volume constant

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

- Avogadro's Law** – The number of gas at constant temperature and pressure is directly proportional to the number of moles of gas.

$$\frac{V_1}{n_1} = \frac{V_2}{n_2}$$

- Combined Gas Law**

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$